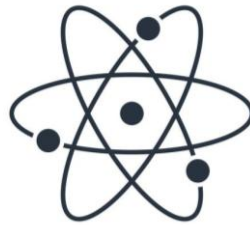


Final Report

Undergraduate Research – CCEN 477

Uncertainty Propagation in Beam-Splitter-Based Quantum Random Number Generators and Its Impact on Uniform and Gaussian Random Number Generation

Supervisor: Dr. Berihu Teklu Gebrehiwot



Team Members:

- Omar Yousef - 100063406
- Ahmad Alhawamdeh - 100063385
- Hasan Al Hussien - 100063390

Date: 22/04/2026

Abstract

Beam-splitter-based quantum random number generators (QRNGs) produce intrinsically unpredictable randomness by exploiting the probabilistic nature of single-photon measurements. The output of such systems can be modeled as a Bernoulli process with parameter p , determined by the effective transmissivity of the beam splitter. In practical implementations, p deviates from the ideal value of 0.5 due to optical losses, detector inefficiencies, and environmental fluctuations, and can only be estimated with finite precision, introducing uncertainty δp into the system.

To address this gap, this work develops a unified analytical framework that links source uncertainty to computational performance across the QRNG pipeline. It is shown that all performance metrics, extraction efficiency, uniform-bit cost, Gaussian generation cost, throughput, and latency are governed by the single function $p(1-p)$, establishing a direct coupling between physical source imperfections and computational overhead. A second-order Taylor expansion yields the approximations $\eta(0.5-\delta p) \approx 0.25 - (\delta p)^2$, $C_u(0.5+\delta p) \approx 4 + 16(\delta p)^2$, and $C_g(0.5+\delta p) \approx 8 + 32(\delta p)^2$, quantifying quadratic degradation in efficiency and cost.

Numerical analysis demonstrates that an uncertainty of $\delta p = 0.10$ increases raw-bit cost by approximately 4%, while a source deviation to $p = 0.2$ increases Gaussian generation cost by over 56% relative to the ideal case. System-level evaluation shows that for a raw measurement rate of 10^6 bits/sec, parameter uncertainty of $\delta p = 0.05$ reduces throughput from 250,000 to as low as 240,000 bits/sec and increases Gaussian sample latency from 8.00 to 8.33 μ s. Classical estimation analysis shows that achieving $\delta p < 0.01$ requires $N \geq 10^4$ samples, introducing a throughput–precision trade-off that is characterized formally. Quantum-enhanced estimation is shown to reduce cost overhead from 1.6% to below 0.01% under the same conditions.

These results establish that randomness generation, extraction, and estimation are intrinsically coupled through the source parameter, and that accurate characterization of p is essential for achieving both statistically correct and computationally efficient QRNG performance.

Table of Contents

1 Introduction	4	5 Classical Estimation of the Source Parameter	16
2 Background and System Model	5	5.1 Bernoulli and Binomial Model	16
2.1 Quantum Random Number Generation	5	5.2 Maximum Likelihood Estimation	16
2.2 Beam-Splitter-Based QRNG Model	5	5.3 Estimation Precision and Fisher Information	17
2.3 Entropy and Randomness Quality	6	5.4 Implications for QRNG Performance	18
2.4 Bias and Randomness Extraction Overview	6	5.5 Sample Size vs Throughput Trade-off	19
2.5 Parameter Estimation and Uncertainty	7	5.6 Bayesian Estimation as an Alternative	19
2.6 Gap in Existing Literature	8	6 System-Level Interpretation	20
2.7 Importance of Uncertainty Propagation	8	6.1 Throughput	20
2.8 Scope and Contributions of This Work	8	6.2 Latency	21
3 Extraction Efficiency Analysis	9	6.3 Effect of Parameter Uncertainty	21
3.1 Von Neumann Extraction Efficiency	9	6.4 Throughput–Precision Trade-off	22
3.2 Sensitivity to the Source Parameter	9	6.5 Failure Mode Analysis	23
3.3 Cost of Uniform Bit Generation	10	6.6 System Performance Comparison	24
3.4 Numerical Evaluation	10	7 Quantum Estimation and Metrology	24
3.5 Comparison with Peres Extractor	11	7.1 Quantum Estimation Framework	25
3.6 Behavior at Extreme Bias	11	7.2 Classical vs Quantum Precision Scaling	25
3.7 Efficiency Under Parameter Uncertainty	12	7.3 Impact on QRNG Performance	26
3.8 Discussion	12	7.4 Practical Limitations	26
4 Cost and Uncertainty Propagation Analysis	12	7.5 Quantum Tomography Perspective	27
4.1 Uniform Random Variables as a Computational Resource	12	7.6 Practical Recommendation	27
4.2 Gaussian Random Number Generation	13	8 Conclusion	27
4.3 Propagation of Parameter Uncertainty	13	References	28
4.4 Extended Numerical Analysis	14		
4.5 System-Level Implications	15		

1. Introduction

Random numbers are a fundamental resource in modern computation, with applications spanning cryptography, secure communications, Monte Carlo simulation, and machine learning. In many of these domains, the quality of randomness affects not only correctness but also performance, reliability, and security [2], [16]. While pseudo-random number generators are widely used due to their efficiency, they remain deterministic and therefore unsuitable for applications requiring true unpredictability.

Quantum random number generators (QRNGs) address this limitation by exploiting the intrinsic probabilistic nature of quantum measurements [1], [2]. In beam-splitter-based QRNGs, a single-photon detection process can be modeled as a Bernoulli random variable with parameter p , which determines both the statistical properties and the efficiency of the generated randomness [4]. In ideal conditions, $p = 0.5$, yielding unbiased outputs. In practice, however, imperfections in optical components and environmental fluctuations introduce deviations and uncertainty in this parameter [2], [3].

Randomness extraction techniques such as the von Neumann method can remove bias and ensure statistically correct output [5]. However, the efficiency of extraction depends strongly on the value of p . As a result, uncertainty in the source parameter does not affect correctness, but directly impacts performance. In particular, it determines the number of raw bits required to generate uniform random bits and propagates to higher-level algorithms such as Gaussian random number generation using transformations like Box–Muller [8], [9].

This report develops a quantitative framework to analyze how uncertainty in the source parameter propagates through the QRNG pipeline. To address this gap, this work develops a unified analytical framework that links source uncertainty to computational performance across the QRNG pipeline. The main contributions are:

- Derivation of a second-order approximation for extraction efficiency
- Development of closed-form bounds under parameter uncertainty
- Quantification of the raw-bit cost of uniform and Gaussian generation
- Numerical evaluation of performance degradation
- System-level interpretation of throughput, latency, and computational cost

The key result is that all performance metrics in the system are governed by the function $p(1 - p)$, which creates a direct coupling between physical imperfections and computational cost. This establishes a unified perspective in which randomness generation, extraction, estimation, and application-level performance are treated as interconnected components of a single system.

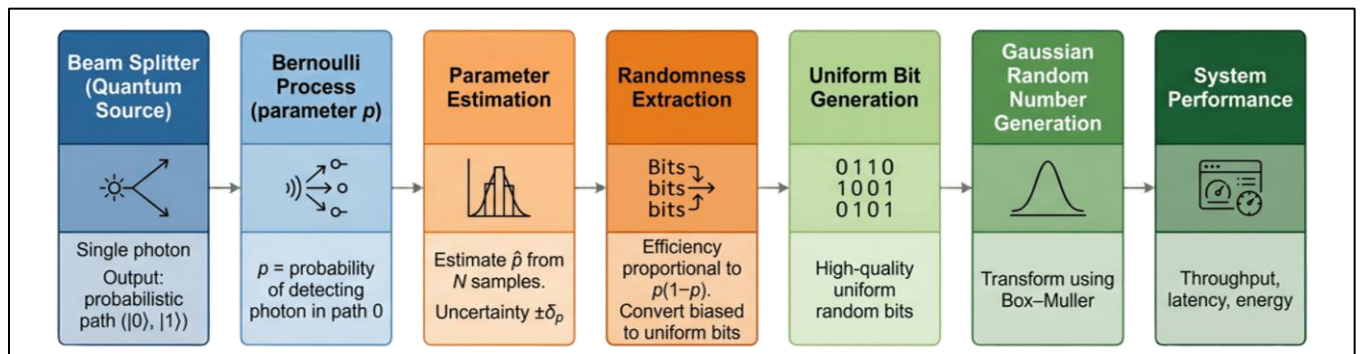


Fig. 1. QRNG processing pipeline and uncertainty propagation.

2. Background and System Model

2.1 Quantum Random Number Generation

Quantum random number generators (QRNGs) exploit the inherent unpredictability of quantum measurement outcomes to generate random numbers. Unlike pseudo-random number generators, which are deterministic algorithms seeded by finite entropy, QRNGs derive randomness from fundamentally indeterministic physical processes [1], [2].

In quantum mechanics, the outcome of a measurement cannot be predicted deterministically even with complete knowledge of the system state. If a system is prepared in a quantum state ρ and measured using a set of operators $\{M_x\}$, the probability of observing outcome x is given by the Born rule:

$$P(X = x) = \text{Tr}(\rho M_x)$$

[1]. This probabilistic mapping defines the entropy source of a QRNG.

In practical implementations, quantum randomness is typically extracted from photonic systems due to their high speed and experimental accessibility [2], [3]. However, the observed measurement statistics are not purely determined by the ideal quantum model. Classical noise, detector inefficiencies, and environmental perturbations modify the effective distribution of outcomes [2], [3].

As a result, the QRNG output cannot be treated as an idealized random source. Instead, it must be modeled as a parameterized stochastic process, where the underlying probability distribution is not perfectly known and may vary over time. This shift in perspective is critical: the problem is no longer only to generate randomness, but to characterize how uncertainty in the physical process propagates through subsequent computational stages.

2.2 Beam-Splitter-Based QRNG Model

A widely studied QRNG architecture is based on a beam splitter. In this system, a single photon incident on a beam splitter is placed in a superposition of transmission and reflection, producing two possible detection outcomes that can be mapped to binary values [1], [4].

This process is modeled as a Bernoulli random variable:

$$P(X = 1) = p, P(X = 0) = 1 - p$$

where p is determined by the effective transmissivity of the beam splitter and the combined effect of optical losses and detector characteristics [4].

In the ideal case, $p = 0.5$, yielding maximum entropy and unbiased randomness. However, real systems deviate from this ideal due to:

- unequal optical path losses,
- detector efficiency mismatch,
- thermal and environmental fluctuations,
- temporal drift in system parameters.

Therefore, the parameter is not fixed but uncertain:

$$p \in [p_0 - \delta p, p_0 + \delta p]$$

This uncertainty model reflects both static imperfections and dynamic fluctuations, and forms the basis for analyzing how performance metrics vary across operating conditions.

2.3 Entropy and Randomness Quality

The amount of randomness generated per measurement is quantified by the Shannon entropy of the Bernoulli distribution:

$$H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

[15].

This function is maximized at $p = 0.5$, where $H(p) = 1$ bit. As p deviates from this value, entropy decreases, indicating a reduction in randomness per measurement.

While entropy is a fundamental measure of randomness quality, it does not capture the efficiency of generating usable random bits. In particular, entropy does not determine how many raw measurements are required to produce unbiased outputs after postprocessing. This distinction becomes important when considering practical systems, where computational cost and throughput are constrained resources.

2.4 Bias and Randomness Extraction

To obtain uniform random bits from a biased source, randomness extraction techniques are employed. The simplest and most widely known method is the von Neumann extractor [5].

This method processes the raw sequence in pairs:

- 01 $\rightarrow$ 0
- 10 $\rightarrow$ 1
- 00, 11 are discarded

The key property is that the output is unbiased regardless of the value of p [5]. However, this comes at the cost of discarding a significant portion of the input data.

The expected extraction efficiency is proportional to:

$$\eta(p) = p(1 - p)$$

which is maximized at $p = 0.5$ and decreases rapidly as the source becomes biased.

More advanced methods, such as the Peres iterative extractor, recover additional randomness from discarded sequences by recursively applying extraction procedures [6]. These methods approach optimal extraction efficiency but introduce additional computational complexity.

From a system perspective, extraction introduces a key transformation:

$$\text{Raw bits} \rightarrow \text{Uniform bits}$$

where the rate of this transformation depends on the source parameter p . Thus, even though extraction guarantees statistical correctness, it introduces a dependency between physical bias and computational cost.

2.5 Parameter Estimation and Uncertainty

In practical QRNG systems, the parameter p is not known a priori and must be estimated from measurement data.

Given N independent samples, the maximum likelihood estimator is:

$$\hat{p} = \frac{1}{N} \sum_{i=1}^N X_i$$

The precision of this estimator is bounded by the Fisher information:

$$I(p) = \frac{1}{p(1-p)}$$

and for N samples:

$$I_N(p) = \frac{N}{p(1-p)}$$

leading to the Cramér–Rao bound:

$$\text{Var}(\hat{p}) \geq \frac{p(1-p)}{N}$$

[10].

These results imply that estimation uncertainty decreases with sample size but cannot be eliminated. Moreover, estimation requires collecting samples that are not used for randomness generation, introducing an inherent trade-off between estimation accuracy and output throughput.

In quantum systems, estimation can also be analyzed using quantum Fisher information, which sets ultimate precision limits and may offer improvements under certain conditions [11], [12], [13].

These results show that parameter estimation and uncertainty can be rigorously quantified using classical statistical tools such as Fisher information and the Cramér–Rao bound [10], as well as their quantum extensions [11]–[13]. However, estimation accuracy is typically analyzed independently of randomness extraction and downstream computational processes. In practical QRNG systems, these components are not isolated. The efficiency of extraction depends on the true value of p , while the estimation process itself introduces uncertainty in that value. This separation in the literature motivates the need for a unified framework that connects source uncertainty, extraction efficiency, and computational cost, which is addressed in the following sections.

2.6 Gap in Existing Literature

Research on QRNGs is typically divided into three domains: physical modeling of entropy sources, randomness extraction algorithms, and statistical estimation theory. These areas are generally treated independently. Extraction methods assume a fixed source distribution, estimation methods do not consider downstream computational cost, and physical models emphasize entropy rather than efficiency.

As a result, there is limited work that integrates these components into a unified framework linking source uncertainty, extraction efficiency, and computational cost. This separation leads to an incomplete understanding of system-level performance.

2.7 Importance of Uncertainty Propagation

The central problem addressed in this work is not bias removal, but the propagation of uncertainty through the QRNG pipeline. Although extraction ensures unbiased output, uncertainty in p leads to variability in extraction efficiency, the number of required raw bits, and the computational cost of downstream algorithms. These effects have direct implications for cryptographic key generation rates, simulation cost, and hardware-level throughput stability.

2.8 Scope and Contributions of This Work

This work develops a unified framework linking source uncertainty to system performance. Specifically, it derives analytical expressions for extraction efficiency as a function of p , establishes closed-form bounds under parameter uncertainty, develops cost models for uniform and Gaussian random number generation, and quantifies performance degradation through numerical analysis. These results enable a system-level interpretation of throughput, latency, and computational cost, providing a direct connection between physical imperfections and computational efficiency.

3. Extraction Efficiency Analysis

In practical QRNG systems, the raw output generated by the quantum source is generally biased due to device imperfections and environmental fluctuations. Consequently, randomness extraction techniques are required to transform the biased Bernoulli sequence into uniform random bits suitable for computational use [2], [5].

While these methods guarantee statistical correctness of the output, their efficiency depends strongly on the underlying source parameter p . As a result, deviations in p do not affect correctness but directly influence the rate at which usable randomness is produced. This section analyzes the extraction efficiency as a function of p , quantifies its sensitivity to deviations from the optimal operating point, and establishes bounds under parameter uncertainty.

3.1 Von Neumann Extraction Efficiency

Consider a Bernoulli source with parameter p , where

$$P(X = 1) = p, P(X = 0) = 1 - p.$$

The von Neumann extractor processes the raw bit stream in pairs and produces an output only when the two bits differ. The probability of producing an output from a pair is:

$$P(\text{output}) = P(01) + P(10) = 2p(1 - p).$$

Since each pair consumes two input bits and produces one output bit when accepted, the extraction efficiency (output bits per input bit) is:

$$\boxed{\eta(p) = p(1 - p)}$$

This function is symmetric around $p = 0.5$ and reaches its maximum:

$$\eta(0.5) = 0.25.$$

3.2 Sensitivity to the Source Parameter

To quantify how efficiency degrades near the optimal operating point, consider a deviation:

$$p = 0.5 - \delta p.$$

Substituting into the efficiency expression:

$$\begin{aligned}\eta(p) &= (0.5 - \delta p)(0.5 + \delta p) \\ \eta(p) &= 0.25 - (\delta p)^2.\end{aligned}$$

Thus,

$$\boxed{\eta(0.5 - \delta p) \approx 0.25 - (\delta p)^2}$$

This result shows that efficiency degradation is quadratic rather than linear, implying that small deviations in p lead to non-negligible performance loss and that sensitivity increases near the optimal operating point. The dependence of extraction efficiency on the source parameter is illustrated in Fig. 2, which shows the parabolic shape of $\eta(p)$ and confirms the quadratic sensitivity derived analytically.

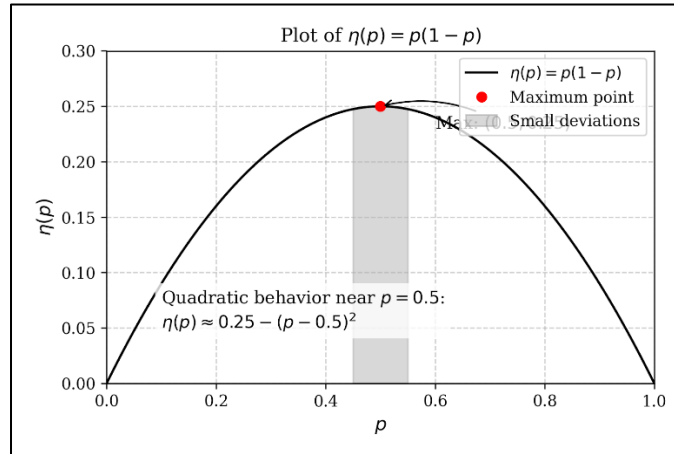


Fig. 2. Extraction efficiency $\eta(p) = p(1-p)$ as a function of the source parameter p , illustrating the quadratic sensitivity to deviations from the optimal operating point $p = 0.5$.

3.3 Cost of Uniform Bit Generation

The cost of generating one unbiased bit is the inverse of efficiency:

$$C_u(p) = \frac{1}{\eta(p)} = \frac{1}{p(1-p)}.$$

At the optimal operating point:

$$C_u(0.5) = 4.$$

Thus, even under ideal conditions, four raw bits are required per output bit on average. As p deviates from 0.5, this cost increases nonlinearly.

3.4 Numerical Evaluation

To quantify this behavior across the operating range, Table 1 presents the extraction efficiency $\eta(p)$ and the corresponding raw-bit cost $C_u(p)$ for representative values of p spanning from the optimal point to strongly biased conditions.

Table 1 - Extraction Efficiency and Cost

p	$\eta(p)$	Raw bits per output C_u
0.50	0.250	4.00
0.48	0.2496	4.01
0.45	0.2475	4.04
0.40	0.2400	4.17
0.35	0.2275	4.40
0.30	0.2100	4.76
0.25	0.1875	5.33

The results demonstrate that:

- Efficiency degrades slowly near $p = 0.5$, but accelerates as bias increases
- A moderate deviation (e.g., $p = 0.3$) increases cost by approximately 19%
- The relationship between bias and cost is strongly nonlinear

3.5 Comparison with Peres Extractor

The von Neumann extractor discards a large fraction of input bits. The Peres extractor improves efficiency by recursively reusing discarded sequences [6].

Unlike the von Neumann method, which achieves efficiency $\eta(p) = p(1 - p)$, the Peres extractor asymptotically approaches the entropy limit:

$$\eta_{\text{Peres}} \approx H(p)$$

where $H(p)$ is the Shannon entropy.

While the Peres method is more efficient, it introduces additional computational complexity. In this work, the von Neumann extractor is used as a baseline because it provides a tractable analytical model for studying the dependence of efficiency on the source parameter.

3.6 Behavior at Extreme Bias

As p approaches 0 or 1:

$$\lim_{p \rightarrow 0,1} \eta(p) = 0$$
$$\lim_{p \rightarrow 0,1} C_u(p) = \infty.$$

This implies that highly biased sources are effectively unusable, as the cost of generating uniform bits diverges.

In practice, the system must operate within a bounded range:

$$p \in [p_{\min}, 1 - p_{\min}]$$

where $p_{\min}$ is determined by acceptable efficiency constraints.

3.7 Efficiency Under Parameter Uncertainty

Given uncertainty:

$$p \in [p_0 - \delta p, p_0 + \delta p],$$

the efficiency lies within:

$$\eta \in [(p_0 - \delta p)(1 - (p_0 - \delta p)), (p_0 + \delta p)(1 - (p_0 + \delta p))].$$

For $p_0 = 0.5$, this simplifies to:

$$\eta_{\min} = 0.25 - (\delta p)^2.$$

Thus, uncertainty in the source parameter translates directly into variability in extraction efficiency and system performance.

3.8 Discussion

The analysis in this section demonstrates that extraction efficiency is entirely determined by the function $p(1 - p)$, which introduces a strong nonlinear dependence on the source parameter. The Taylor expansion shows that efficiency degradation is quadratic near the optimal operating point, implying that even small deviations in p lead to measurable performance loss. This effect becomes more pronounced as the source becomes increasingly biased, with the cost of generating uniform bits growing rapidly and diverging in the extreme case.

This behavior is consistent with the theoretical properties of randomness extraction, where efficiency is fundamentally limited by the entropy of the source and the structure of the extraction algorithm [6], [7]. Furthermore, when parameter uncertainty is taken into account, efficiency is no longer a single value but an interval, introducing variability in system performance. These results establish extraction efficiency as the primary mechanism through which physical imperfections propagate to computational cost.

4. Cost and Uncertainty Propagation Analysis

4.1 Uniform Random Variables as a Computational Resource

The extraction efficiency derived in Section 3 establishes a direct relationship between the source parameter p and the rate at which usable randomness is produced. Since the von Neumann extractor produces unbiased bits with efficiency

$$\eta(p) = p(1 - p),$$

the expected number of raw bits required to generate one uniform random bit is

$$C_u(p) = \frac{1}{\eta(p)} = \frac{1}{p(1 - p)}.$$

This expression defines the computational cost of uniform randomness, measured in raw quantum measurements per output bit.

To connect this cost model with the sensitivity analysis developed in Section 3.2, consider a deviation from the optimal operating point:

$$p = 0.5 + \delta p.$$

Substituting into the cost function:

$$C_u(p) = \frac{1}{(0.5 + \delta p)(0.5 - \delta p)} = \frac{1}{0.25 - (\delta p)^2}.$$

For small δp , we expand using a Taylor approximation:

$$\frac{1}{0.25 - x} \approx \frac{1}{0.25} + \frac{x}{0.25^2}$$

with $x = (\delta p)^2$, yielding:

$$C_u(0.5 + \delta p) \approx 4 + 16(\delta p)^2.$$

This result shows that the cost increases quadratically with deviations in p , reinforcing the sensitivity observed in the efficiency function. Although this approximation is derived around $p = 0.5$, it remains accurate for small deviations near this operating region, including practical values such as $p_0 = 0.48$.

This approximation is accurate for small deviations ($|\delta p| < 0.1$), but becomes less precise for strongly biased sources, where exact expressions should be used.

4.2 Gaussian Random Number Generation

Uniform random bits are typically not the final output in many applications. Instead, they serve as inputs to higher-level transformations that generate random variables with specific distributions. One of the most common examples is Gaussian random number generation using the Box–Muller transform [8], [9]:

$$Z_1 = \sqrt{-2\ln U_1} \cos(2\pi U_2), Z_2 = \sqrt{-2\ln U_1} \sin(2\pi U_2),$$

where U_1 and U_2 are independent uniform random variables.

Each pair (U_1, U_2) produces two Gaussian samples. Therefore, generating one Gaussian random variable requires, on average, one uniform random variable, which itself is produced from extracted random bits.

Combining this with the cost of uniform generation:

$$C_g(p) = 2 \cdot C_u(p) = \frac{2}{p(1-p)}.$$

At the optimal operating point:

$$C_g(0.5) = 8.$$

Thus, even under ideal conditions, generating a single Gaussian random variable requires an average of 8 raw bits.

Using the same approximation as before:

$$C_g(0.5 + \delta p) = 2C_u(0.5 + \delta p) \approx 8 + 32(\delta p)^2.$$

This shows that Gaussian generation amplifies the cost sensitivity introduced at the extraction stage.

4.3 Propagation of Parameter Uncertainty

To quantify the impact of uncertainty in p , consider:

$$p \in [p_0 - \delta p, p_0 + \delta p]$$

For $p_0 = 0.5$, efficiency is maximized at the center, so:

$$\eta_{\max} = 0.25, \eta_{\min} = 0.25 - (\delta p)^2.$$

From this, the worst-case cost is:

$$C_{u,\max} = \frac{1}{\eta_{\min}}, C_{g,\max} = \frac{2}{\eta_{\min}}.$$

Table 2 summarizes the worst-case efficiency, cost, and percentage overhead for increasing values of δp , with p_0 fixed at 0.5.

Table 2 - Uncertainty Propagation

δp	$\eta_{\min}$	$C_{u,\max}$	$C_{g,\max}$	Overhead (%)
0.00	0.2500	4.00	8.00	0.0
0.02	0.2496	4.01	8.02	0.4
0.05	0.2475	4.04	8.08	1.6
0.08	0.2436	4.10	8.21	2.5
0.10	0.2400	4.17	8.33	4.2
0.12	0.2356	4.25	8.50	6.3
0.15	0.2275	4.40	8.80	10.0

The results demonstrate that cost increases nonlinearly with uncertainty. For $\delta p = 0.10$, the system experiences over 4% overhead, while for $\delta p = 0.15$, the cost increases by approximately 10%. These values indicate that even moderate uncertainty can significantly degrade system performance. The overall uncertainty propagation chain is illustrated in Fig. 3

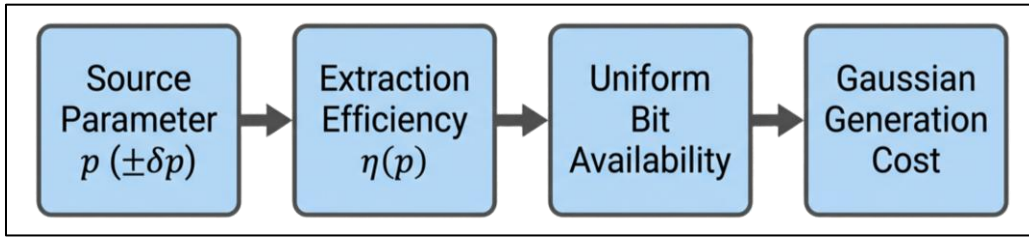


Fig. 3. Propagation of uncertainty from the source parameter to computational cost.

4.4 Extended Numerical Analysis

To evaluate performance across a wider operating range, consider the following values of p :

Table 3 - Efficiency and Cost vs p

p	$\eta(p)$	C_u	C_g	Overhead (%)
0.50	0.250	4.00	8.00	0.0
0.48	0.2496	4.01	8.02	0.3
0.45	0.2475	4.04	8.08	1.0
0.42	0.2436	4.10	8.21	2.5
0.40	0.2400	4.17	8.33	4.2
0.35	0.2275	4.40	8.80	10.0
0.30	0.2100	4.76	9.52	19.0
0.25	0.1875	5.33	10.67	33.3
0.20	0.1600	6.25	12.50	56.3

The results show that cost increases slowly near $p = 0.5$, but accelerates rapidly as the source becomes biased. At $p = 0.3$, the Gaussian generation cost increases by approximately 19%, and at $p = 0.2$, the cost increases by more than 50%, indicating a severe loss of efficiency.

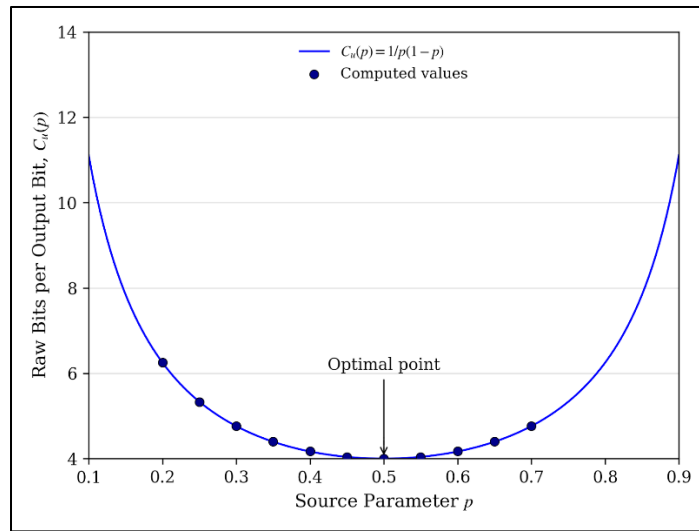


Fig. 4. Raw-bit cost $C_u(p) = 1/p(1 - p)$ vs. source parameter p , showing minimum at $p = 0.5$ and nonlinear increase with bias.

4.5 System-Level Implications

To quantify the system-level impact of parameter uncertainty, consider a QRNG operating at a raw measurement rate

$$R = 10^6 \text{ bits/sec}$$

with nominal parameter

$$p_0 = 0.48, \delta p = 0.03.$$

The efficiency bounds are:

$$\begin{aligned} \eta_{\min} &= (0.48 - 0.03)(1 - (0.48 - 0.03)) = 0.45 \times 0.55 = 0.2475, \\ \eta_{\max} &= 0.48 \times 0.52 = 0.2496. \end{aligned}$$

The corresponding throughput range is:

$$T \in [R \cdot \eta_{\min}, R \cdot \eta_{\max}] = [247500, 249600] \text{ bits/sec.}$$

Since Gaussian generation requires two uniform bits per sample, the Gaussian output rate is:

$$T_g \in [123750, 124800] \text{ samples/sec (Gaussian output rate)}$$

This corresponds to approximately a $\pm 0.4\%$ variation in output rate. Although this variation appears small in relative terms, it reflects a fundamental property of the system: the efficiency of randomness generation is directly tied to the source parameter, and any uncertainty in that parameter propagates into measurable variability in system performance.

As the source deviates from the optimal operating point $p = 0.5$, the extraction efficiency decreases, resulting in fewer usable bits per unit time and a corresponding reduction in throughput. At the same time, the number of required raw measurements increases, leading to higher computational cost and increased latency. This effect becomes more pronounced when scaled to high-throughput systems, where even small percentage changes translate into significant absolute differences in output rate.

Moreover, when the parameter p fluctuates within an interval rather than remaining fixed, the system no longer operates at a constant efficiency. Instead, both throughput and computational cost vary over time, introducing unpredictability into the system. This variability can complicate system design, particularly in applications requiring deterministic performance or strict timing guarantees.

These effects are especially relevant in high-performance and security-critical applications. Variations in randomness generation may impact resource allocation, energy consumption, and buffering requirements. Consequently, QRNG systems must be analyzed not only in terms of statistical correctness but also in terms of performance stability under parameter uncertainty.

5. Classical Estimation of the Source Parameter

In practical QRNG systems, the parameter p of the Bernoulli source is not known a priori and must be estimated from observed measurement data. Accurate estimation of p is essential, as uncertainty in this parameter directly affects randomness extraction efficiency and the performance of downstream processes [10].

5.1 Bernoulli and Binomial Model

The output of a beam-splitter-based QRNG can be modeled as a sequence of independent and identically distributed Bernoulli random variables with parameter p [1], [4]. Each measurement produces a binary outcome corresponding to photon detection in one of two possible paths. For a sequence of N measurements, the total number of occurrences of one outcome follows a binomial distribution:

$$K \sim \text{Binomial}(N, p).$$

This probabilistic model provides the basis for estimating the unknown parameter p from observed data.

5.2 Maximum Likelihood Estimation

Given observations $X_1, X_2, \dots, X_N$, the maximum likelihood estimator (MLE) for p is:

$$\hat{p} = \frac{1}{N} \sum_{i=1}^N X_i.$$

This estimator is unbiased and consistent, meaning it converges to the true parameter as $N \rightarrow \infty$ [10]. However, the estimator's accuracy depends on the number of samples collected, which introduces a trade-off between estimation precision and system throughput.

5.3 Estimation Precision and Fisher Information

The precision of the estimator is fundamentally limited by the Fisher information of the Bernoulli distribution:

$$I(p) = \frac{1}{p(1-p)}$$

[10]. For N independent samples, the total Fisher information is:

$$I_N(p) = \frac{N}{p(1-p)}.$$

The Cramér–Rao bound gives the minimum achievable variance:

$$\text{Var}(\hat{p}) \geq \frac{p(1-p)}{N}$$

[10]. The corresponding standard deviation is:

$$\sigma = \sqrt{\frac{p(1-p)}{N}}.$$

A 95% confidence interval is given by:

$$\hat{p} \pm 1.96\sigma.$$

Table 4 - Estimation Precision for $p = 0.5$

N	Var($\hat{p}$)	σ	95% CI Half-Width
100	0.0025	0.0500	± 0.098
500	0.0005	0.0224	± 0.0439
1000	0.00025	0.0158	± 0.0310
5000	0.00005	0.0071	± 0.0139
10000	0.000025	0.0050	± 0.0098

The dependence of estimation uncertainty on the source parameter is illustrated in Fig. 5.

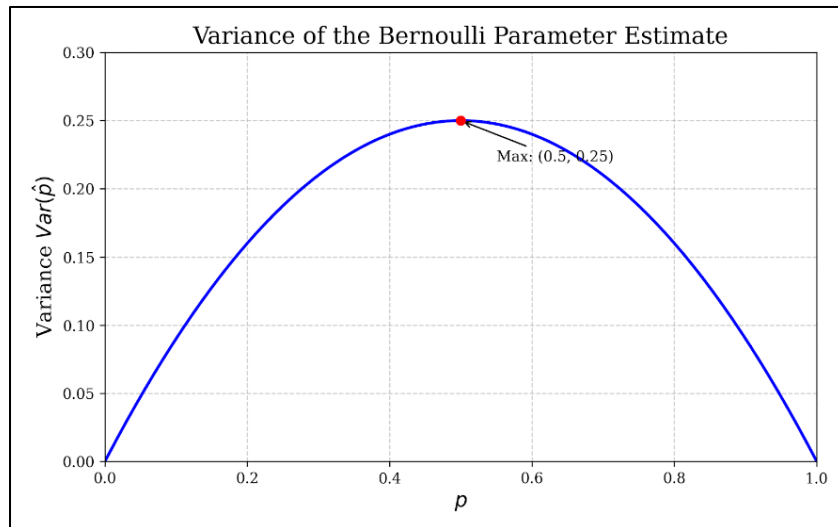


Fig. 5. Estimation variance $\text{Var}(\hat{p}) = \frac{p(1-p)}{N}$ as a function of p .

Fig. 5 plots this variance as a continuous function of p for a fixed sample size, illustrating that uncertainty is highest near $p = 0.5$ where the source is maximally random.

For a biased source $p = 0.35$, the variance becomes:

$$\text{Var}(\hat{p}) = \frac{0.35 \cdot 0.65}{N} = \frac{0.2275}{N}.$$

Table 5 - Estimation Precision for $p = 0.35$

N	Var($\hat{p}$)	σ	95% CI Half-Width
100	0.002275	0.0477	± 0.0935
500	0.000455	0.0213	± 0.0418
1000	0.0002275	0.0151	± 0.0296
5000	0.0000455	0.0067	± 0.0131
10000	0.00002275	0.0048	± 0.0094

These results show that estimation precision depends on both sample size and the value of p . The variance is maximized at $p = 0.5$, indicating that estimating an unbiased source is statistically more challenging than estimating a biased one.

5.4 Implications for QRNG Performance

The estimation results demonstrate that achieving high precision requires a large number of samples. For example, reducing the estimation uncertainty to approximately ± 0.01 around $p = 0.5$ requires on the order of 10^4 samples. Since these samples must be collected before accurate system operation can be guaranteed, estimation introduces a delay during which no usable output is produced.

This creates a direct trade-off between estimation accuracy and randomness generation throughput. Increasing N improves parameter precision but reduces the effective rate of usable randomness, particularly in systems where estimation must be performed repeatedly due to parameter drift.

5.5 Sample Size vs Throughput Trade-off

During the estimation phase, raw bits are consumed but not used for randomness generation. Let f denote the fraction of time spent estimating. The effective throughput is:

$$T_{\text{eff}} = (1 - f) \cdot R \cdot \eta(p).$$

This expression shows that throughput decreases linearly with the estimation fraction f . Increasing the sample size N improves estimation accuracy (reducing δp) but increases f , thereby reducing effective throughput.

The optimal value of f depends on the trade-off between precision and efficiency. In systems with rapidly varying parameters, smaller N may be preferred to maintain throughput, while in stable systems, larger N can be used to reduce uncertainty and improve efficiency in the long term.

5.6 Bayesian Estimation as an Alternative

An alternative to batch estimation is Bayesian inference. For a Bernoulli process, the conjugate prior is the Beta distribution [10]. If the prior is:

$$p \sim \text{Beta}(\alpha, \beta),$$

and K successes are observed in N trials, the posterior distribution becomes:

$$p \sim \text{Beta}(\alpha + K, \beta + N - K).$$

The posterior mean is:

$$E[p] = \frac{\alpha + K}{\alpha + \beta + N}.$$

This approach allows sequential updating of the parameter estimate without requiring fixed sample sizes. It is particularly useful for tracking time-varying parameters, as it continuously incorporates new observations.

Compared to MLE, Bayesian estimation provides smoother updates and incorporates prior information, but it introduces additional computational complexity and depends on the choice of prior distribution.

From a system perspective, Bayesian estimation reduces fluctuations in the estimate of p over time, which can stabilize extraction efficiency and reduce variability in computational cost, at the expense of increased computational complexity.

For example, compared to MLE, Bayesian estimation produces smoother updates in p for small sample sizes, reducing short-term fluctuations in estimated efficiency.

6. System-Level Interpretation

The results developed in the previous sections establish a direct connection between the physical properties of the QRNG source and the performance of higher-level computational processes. In particular, uncertainty in the source parameter p propagates through the randomness generation pipeline and affects system-level behavior in a measurable way.

From Sections 3 and 4, the efficiency of randomness extraction and the cost of generating uniform and Gaussian random variables are governed by the function $p(1 - p)$. As shown in the numerical analysis, deviations from the optimal value $p = 0.5$ lead to an increase in the number of required raw bits. For example, a shift from $p = 0.5$ to $p = 0.3$ increases the cost from 4 to approximately 4.76 raw bits per output bit, representing a non-negligible increase in computational effort.

6.1 Throughput

The throughput of a QRNG system is defined as the rate at which usable uniform random bits are produced. Given a raw measurement rate R (in bits per second), the effective throughput is determined by the extraction efficiency:

$$T(p) = R \cdot \eta(p) = R \cdot p(1 - p).$$

This relationship follows directly from the extraction model developed in Section 3 and the cost analysis in Section 4.

For a representative system with:

$$R = 10^6 \text{ bits/sec},$$

the throughput for different values of p is given in Table 6.

Table 6 - Throughput vs Source Parameter

p	$\eta(p)$	Throughput (bits/sec)	Degradation (%)
0.50	0.250	250,000	0.0
0.45	0.2475	247,500	1.0
0.40	0.2400	240,000	4.0
0.35	0.2275	227,500	9.0
0.30	0.2100	210,000	16.0

(Assuming raw measurement rate $R = 10^6$ bits/sec)

The results show that throughput decreases nonlinearly as the source becomes biased. Even moderate deviations from $p = 0.5$ result in measurable degradation, which becomes significant at larger bias values. This behavior reflects the dependence of extraction efficiency on $p(1 - p)$, as derived previously.

6.2 Latency

Latency is defined as the expected time required to generate one Gaussian random sample. From Section 4, the cost of generating one Gaussian sample is:

$$C_g(p) = \frac{2}{p(1 - p)}.$$

Given a raw bit rate R , the latency is:

$$L(p) = \frac{C_g(p)}{R} = \frac{2}{R \cdot p(1 - p)}.$$

For $R = 10^6$ bits/sec, the latency values are:

Table 7 - Latency vs Source Parameter

p	$C_g(p)$	Latency (μ s)
0.50	8.00	8.00
0.45	8.08	8.08
0.40	8.33	8.33
0.35	8.80	8.80
0.30	9.52	9.52

(Assuming raw measurement rate $R = 10^6$ bits/sec)

These results show that latency increases as the source becomes biased. Although the increase appears gradual, it reflects the same nonlinear dependence observed in throughput and cost.

In real-time systems, even small increases in latency can accumulate over large numbers of samples, leading to performance bottlenecks.

6.3 Effect of Parameter Uncertainty

In practical systems, the parameter p is not fixed but lies within an uncertainty interval:

$$p \in [p_0 - \delta p, p_0 + \delta p].$$

Case 1: $p_0 = 0.48, \delta p = 0.03$

$$p \in [0.45, 0.51]$$

Efficiency bounds:

$$\eta_{\min} = 0.45 \times 0.55 = 0.2475, \eta_{\max} = 0.48 \times 0.52 = 0.2496$$

Throughput:

$$T \in [247,500, 249,600] \text{ bits/sec}$$

Latency:

$$L \in [8.02, 8.08] \mu\text{s}$$

Case 2: $p_0 = 0.45, \delta p = 0.05$

$$p \in [0.40, 0.50]$$

$$\eta_{\min} = 0.40 \times 0.60 = 0.24, \eta_{\max} = 0.45 \times 0.55 = 0.2475$$

$$T \in [240,000, 247,500] \text{ bits/sec}$$

$$L \in [8.08, 8.33] \mu\text{s}$$

These results show that uncertainty introduces a range of possible system behaviors rather than a single deterministic performance value. The width of this range increases with both bias and uncertainty, reducing predictability.

6.4 Throughput–Precision Trade-off

In practical QRNG systems, parameter estimation and randomness generation are not independent processes. Estimation requires collecting samples from the source, during which no usable output is produced. This introduces a trade-off between estimation accuracy and system throughput.

Let f denote the fraction of time spent performing parameter estimation. The effective throughput is then:

$$T_{\text{eff}} = (1 - f) R \cdot \eta(p),$$

where R is the raw measurement rate and $\eta(p)$ is the extraction efficiency derived in Section 3.

From Section 5, achieving a desired estimation precision requires a sample size N , which determines the duration of the estimation phase. Increasing N reduces the estimation error δp , as governed by the Cramér–Rao bound [10], but increases f , thereby reducing throughput.

At the same time, reducing δp improves efficiency, since:

$$\eta(0.5 \pm \delta p) \approx 0.25 - (\delta p)^2.$$

Thus, there are two competing effects:

- Increasing $N \rightarrow$ decreases $\delta p \rightarrow$ improves efficiency
- Increasing $N \rightarrow$ increases $f \rightarrow$ reduces throughput

To identify the optimal operating point, consider that throughput depends on both factors:

$$T_{\text{eff}} \propto (1 - f)(0.25 - (\delta p)^2).$$

This trade-off implies the existence of an optimal sample size N that maximizes throughput by balancing the quadratic improvement in efficiency with the linear reduction in available output time.

In systems where p is relatively stable, larger values of N can be used to reduce uncertainty and improve long-term efficiency. In contrast, for time-varying systems, smaller N may be preferred to reduce estimation overhead and maintain consistent output rates.

This trade-off highlights that estimation is not a purely statistical problem but a system-level design decision that directly affects performance.

The relationship between effective throughput and parameter deviation for varying estimation overhead fractions is shown in Fig. 6.

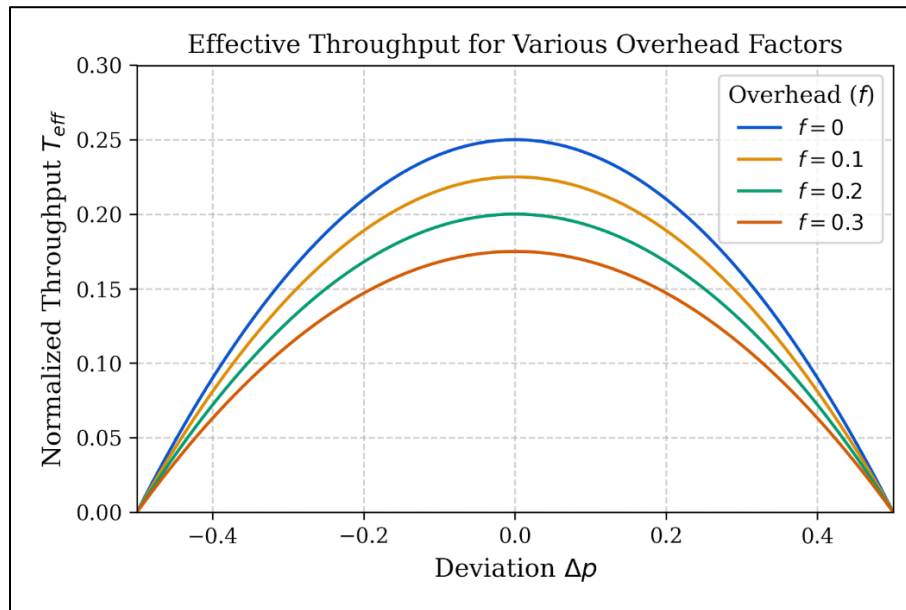


Fig. 6. Effective throughput T_{eff} vs. parameter deviation Δp for different estimation overhead fractions f .

6.5 Failure Mode Analysis

In practical applications, systems often require a minimum guaranteed throughput $T_{\min}$. This defines a constraint:

$$T(p) = R \cdot p(1 - p) \geq T_{\min}.$$

Solving:

$$p(1 - p) = \frac{T_{\min}}{R}.$$

This quadratic equation defines critical values p^* such that:

$$p^*(1 - p^*) = \frac{T_{\min}}{R}.$$

The system operates safely only if:

$$p \in [p^*, 1 - p^*].$$

If the uncertainty interval:

$$[p_0 - \delta p, p_0 + \delta p]$$

extends outside this range, the system cannot guarantee minimum throughput.

This defines a failure condition driven entirely by parameter uncertainty.

6.6 System Performance Comparison

To summarize system behavior under different conditions, consider three scenarios.

Table 8 - System Performance Comparison

Scenario	η Range	Cu Range	Cg Range	Throughput (bits/sec)	Latency (μ s)	Stability
Ideal (0.5,0)	[0.25, 0.25]	[4.00, 4.00]	[8.00, 8.00]	250k	8.00	Stable
Moderate (0.48,0.02)	[0.2496, 0.25]	[4.00, 4.01]	[8.00, 8.02]	249k–250k	8.00–8.02	Stable
Degraded (0.45,0.05)	[0.24, 0.2475]	[4.04, 4.17]	[8.08, 8.33]	240k–247k	8.08–8.33	Marginal

This comparison highlights how increasing bias and uncertainty degrade both efficiency and stability. The throughput and latency ranges across all three scenarios are compared visually in Fig. 7.

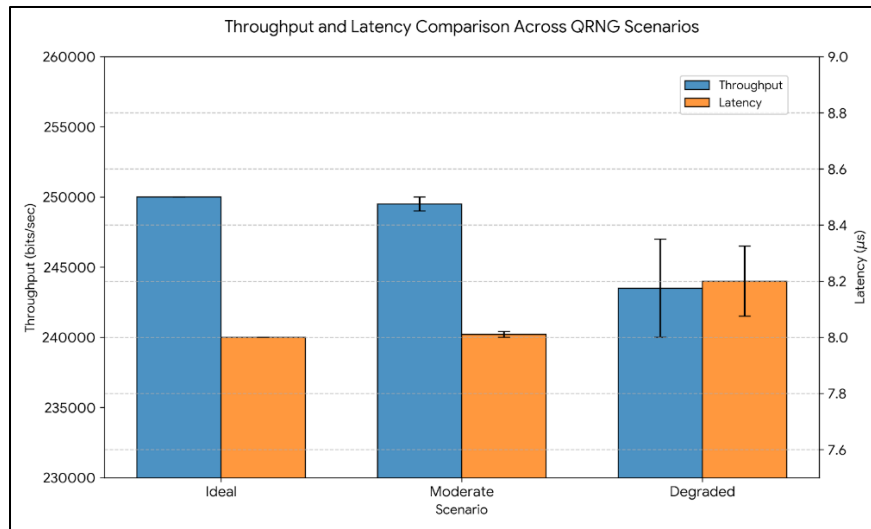


Fig. 7. Throughput (bits/sec) and latency (μs) comparison across QRNG operating scenarios (separate axes).

Note: Latency scale is inverted for visualization

From a practical perspective, the magnitude of the observed overhead can be compared to typical QRNG implementations reported in the literature, where throughput variations of a few percent are common due to hardware imperfections and environmental fluctuations. The cost increases predicted by the model are therefore consistent with expected system-level behavior. However, real optical QRNG systems may exhibit additional inefficiencies due to detector noise and finite bandwidth, which are not captured in the current framework. This highlights that while the analytical model provides useful performance bounds, actual implementations may experience larger variability.

7. Quantum Estimation and Metrology

The limitations of classical estimation methods motivate the use of quantum estimation techniques, which aim to improve parameter estimation by exploiting the properties of quantum systems. In beam-splitter-based QRNGs, the parameter of interest is the transmissivity τ , which directly determines the Bernoulli parameter p , such that:

$$p = \tau$$

Thus, estimating p can be formulated as a quantum parameter estimation problem [11].

7.1 Quantum Estimation Framework

In classical estimation, repeated independent measurements reduce uncertainty according to the standard quantum limit (SQL), where the estimation error scales as $\delta\lambda \sim 1/\sqrt{N}$. This scaling arises from averaging independent outcomes. In contrast, quantum estimation strategies exploit correlations such as entanglement, enabling Heisenberg-limited scaling of $\delta\lambda \sim 1/N$, which represents a fundamentally different regime of precision rather than a simple improvement in constant factors.

$$\delta\lambda_{\text{SQL}} = \frac{1}{\sqrt{N}},$$

which is known as the standard quantum limit (SQL) [12]. In contrast, quantum estimation techniques can exploit non-classical resources such as entanglement to achieve improved scaling, approaching the Heisenberg limit:

$$\delta\lambda_{\text{HL}} = \frac{1}{N},$$

which represents a quadratic improvement in precision [11], [12].

Quantum estimation theory extends classical estimation by incorporating the quantum state of the system and the measurement process. The ultimate precision is bounded by the quantum Cramér–Rao bound:

$$\text{Var}(\hat{\lambda}) \geq \frac{1}{NF_Q(\lambda)},$$

where $F_Q(\lambda)$ is the quantum Fisher information [11]. This bound defines the fundamental limit on how accurately parameters such as the beam splitter transmissivity can be estimated.

In the context of the QRNG model developed in this work, improved estimation precision directly reduces uncertainty in p , which in turn reduces variability in extraction efficiency and computational cost as quantified in Sections 4 and 6.

7.2 Classical vs Quantum Precision Scaling

To compare classical and quantum estimation performance, consider the estimation error as a function of the number of measurements N :

- Standard Quantum Limit (SQL):

$$\delta\lambda_{\text{SQL}} = \frac{1}{\sqrt{N}}$$

- Heisenberg Limit:

$$\delta\lambda_{\text{HL}} = \frac{1}{N}$$

[12].

Table 9 - Precision Scaling Comparison

N	$\delta\lambda_{\text{SQL}}$	$\delta\lambda_{\text{HL}}$	Improvement Factor
10^2	0.100	0.010	10
10^3	0.0316	0.0010	31.6
10^4	0.0100	0.0001	100
10^5	0.00316	0.00001	316

These results show that quantum-enhanced estimation can significantly reduce estimation error, particularly at large N . For example, at $N = 1000$, classical estimation yields an uncertainty of approximately 0.0316, whereas quantum-enhanced estimation reduces this to 0.001, representing an improvement by more than an order of magnitude.

7.3 Impact on QRNG Performance

The improvement in estimation precision has direct consequences for the system-level performance analyzed in Section 4.

From Section 5, classical estimation with $N = 1000$ yields approximately:

$$\delta p \approx 0.05.$$

Using Table 2 from Section 4.3, this corresponds to:

$$C_u \approx 4.04, C_g \approx 8.08.$$

If quantum-enhanced estimation reduces the uncertainty to:

$$\delta p \approx 0.005,$$

then:

$$\eta_{\min} = 0.25 - (0.005)^2 = 0.249975, \\ C_u \approx 4.0004, C_g \approx 8.0008.$$

Thus, the cost overhead is reduced from approximately 1.6% to less than 0.01%, effectively restoring near-ideal system performance.

This demonstrates that improved parameter estimation directly translates into reduced computational cost and improved throughput stability.

7.4 Practical Limitations

Despite the theoretical advantages of quantum-enhanced estimation, practical implementation presents significant challenges. Achieving Heisenberg-limited scaling requires the use of entangled or highly non-classical states, which are sensitive to noise, loss, and decoherence [12], [13].

In optical QRNG systems, photon loss and detector inefficiencies degrade the achievable precision, often limiting performance to near-SQL scaling. Additionally, generating and maintaining entangled states at high rates is experimentally demanding and may offset the gains in estimation accuracy.

Therefore, while quantum estimation provides a theoretical advantage, its practical applicability in QRNG systems is constrained by implementation complexity and noise.

7.5 Quantum Tomography Perspective

An alternative approach to parameter estimation is quantum process tomography, which seeks to reconstruct the full transformation performed by the beam splitter rather than estimating a single parameter. This provides a more complete characterization of the system, including loss, phase shifts, and other imperfections.

However, tomography requires a large number of measurements and extensive postprocessing, making it computationally expensive. For practical QRNG systems, where real-time performance is required, tomography is typically not feasible as a continuous monitoring method.

Instead, it may be used for initial calibration or validation of the system.

7.6 Practical Recommendation

Based on the analysis in this section and previous sections, the choice of estimation strategy involves a trade-off between precision, complexity, and system performance.

Classical maximum likelihood estimation (MLE) with sufficiently large sample size remains the most practical approach for most QRNG systems. It is simple to implement, computationally efficient, and provides acceptable accuracy when N is large [10]. However, it introduces a trade-off between estimation accuracy and throughput, as discussed in Section 5.

Bayesian estimation provides a more flexible alternative, allowing sequential updates and adaptation to time-varying parameters. This is particularly useful in systems where p drifts over time, as it avoids the need for fixed batch sizes and enables continuous tracking of the parameter.

Quantum-enhanced estimation offers the potential for significantly improved precision, reducing uncertainty and stabilizing system performance. However, its practical implementation is limited by experimental constraints, including noise, decoherence, and the difficulty of generating entangled states at scale [11]–[13].

In the context of beam-splitter-based QRNG systems, the most practical approach is a hybrid strategy combining classical estimation with adaptive or Bayesian updating. This approach provides a balance between accuracy and efficiency while remaining feasible for real-time implementation.

7.7 Limitations

This work is based on a simplified analytical model of a beam-splitter-based QRNG and therefore has several limitations. First, the analysis assumes a single-photon Bernoulli source and does not account for multi-photon effects or more complex quantum optical behavior. Second, detector imperfections such as dark counts, dead time, and temporal correlations are not explicitly modeled, although they may influence the effective value of the parameter p in practical systems.

In addition, the analysis is purely theoretical and does not include experimental validation. Real-world QRNG implementations may exhibit additional sources of noise and non-idealities that are not captured in the model. Finally, the framework assumes independent and identically distributed samples, which may not hold under hardware drift or correlated noise conditions.

These limitations indicate that the results should be interpreted as idealized performance bounds rather than exact predictions for specific physical implementations.

8. Conclusion

The central problem addressed in this work is the impact of uncertainty in the source parameter p of a beam-splitter-based quantum random number generator on system performance. While randomness extraction techniques ensure that the output is statistically unbiased, they do not guarantee consistent efficiency. This report showed that extraction efficiency, computational cost, and system-level performance are all governed by the function $p(1 - p)$, establishing a direct coupling between physical source imperfections and computational overhead.

The analytical results demonstrate that deviations from the optimal operating point $p = 0.5$ lead to quadratic degradation in extraction efficiency. Specifically, the approximation

$$\eta(0.5 - \delta p) \approx 0.25 - (\delta p)^2$$

shows that even small deviations produce measurable losses. Correspondingly, the cost of generating uniform random bits follows

$$C_u(0.5 + \delta p) \approx 4 + 16(\delta p)^2,$$

while the cost of generating Gaussian random variables increases as

$$C_g(0.5 + \delta p) \approx 8 + 32(\delta p)^2.$$

The numerical analysis confirms these results. For example, an uncertainty of $\delta p = 0.10$ increases the raw-bit cost by approximately 4%, while $\delta p = 0.15$ leads to a cost increase of approximately 10%. Similarly, a shift from $p = 0.5$ to $p = 0.3$ increases the Gaussian generation cost from 8 to approximately 9.52 raw bits per sample. These results highlight the nonlinear sensitivity of the system to bias and uncertainty.

The estimation analysis further shows that achieving high precision in p requires a large number of samples. For instance, reducing estimation uncertainty to approximately ± 0.01 at $p = 0.5$ requires on the order of $N \geq 10^4$ samples based on the Cramér–Rao bound [10]. This introduces a trade-off between estimation accuracy and throughput, since data used for estimation cannot be used for randomness generation. Bayesian estimation provides a flexible alternative by enabling sequential updates without fixed batch sizes [10], while quantum-enhanced estimation offers improved precision scaling at the cost of increased experimental complexity [11]–[13].

These findings show that QRNG performance must be analyzed as a coupled system involving physical randomness generation, parameter estimation, randomness extraction, and computational cost. Treating these components independently, as is commonly done in the literature [1], [2], [7], leads to an incomplete understanding of system behavior. A unified framework is required to capture the propagation of uncertainty across all stages.

This study identifies four concrete directions for future work. First, experimental validation of the theoretical results using a physical beam-splitter-based QRNG would provide practical confirmation of the predicted cost and efficiency behavior. Second, implementing adaptive estimation methods, such as Bayesian updating, would allow real-time tracking of parameter variations and improved system stability. Third, extending the framework to more complex QRNG architectures, including multi-photon and entangled-state systems, would enable analysis of next-generation quantum randomness sources. Finally, integrating the model into a full cryptographic protocol, such as quantum key distribution, would allow evaluation of end-to-end system performance under realistic uncertainty conditions.

In conclusion, this work establishes that uncertainty in the physical source parameter is not merely a statistical detail but a primary driver of computational cost and system performance. The framework developed here is applicable not only to QRNGs but to any physical random number generation system in which uncertainty in the underlying process propagates to algorithmic efficiency and resource consumption.

References

- [1] X. Ma, X. Yuan, Z. Cao, B. Qi, and Z. Zhang, “Quantum random number generation,” *arXiv preprint arXiv:1510.08957*, 2016.
- [2] M. Herrero-Collantes and J. C. Garcia-Escartin, “Quantum random number generators,” *arXiv preprint arXiv:1604.03304*, 2016.
- [3] Z. Haider et al., “Quantum Random Number Generator (QRNG): Theoretical and Experimental Investigations,” *arXiv preprint arXiv:2506.02441*, 2025.

- [4] O. Yousef et al., “Precise characterization of a quantum random source and its impact on uniform and Gaussian random number generation,” Undergraduate Research Literature Review, Khalifa University, 2026.
- [5] J. von Neumann, “Various techniques used in connection with random digits,” *Applied Math Series*, 1951.
- [6] Y. Peres, “Iterating von Neumann’s procedure for extracting random bits,” *Annals of Statistics*, vol. 20, no. 1, pp. 590–597, 1992.
- [7] N. Nisan and A. Ta-Shma, “Extracting randomness: A survey and new constructions,” *Journal of Computer and System Sciences*, vol. 58, no. 1, pp. 148–173, 1999.
- [8] G. E. P. Box and M. E. Muller, “A note on the generation of random normal deviates,” *Annals of Mathematical Statistics*, vol. 29, no. 2, pp. 610–611, 1958.
- [9] L. Devroye, *Non-Uniform Random Variate Generation*. New York, NY, USA: Springer, 1986.
- [10] A. Ly, M. Marsman, J. Verhagen, R. P. P. P. Grasman, and E.-J. Wagenmakers, “A tutorial on Fisher information,” *Journal of Mathematical Psychology*, vol. 80, pp. 40–55, 2017.
- [11] M. G. A. Paris,
“Quantum estimation for quantum technology,” *arXiv preprint arXiv:0804.2981*, 2009.
- [12] V. Giovannetti, S. Lloyd, and L. Maccone, “Advances in quantum metrology,” *arXiv preprint arXiv:1102.2318*, 2011.
- [13] A. Monras and M. G. A. Paris, “Optimal quantum estimation of loss in bosonic channels,” *Physical Review Letters*, 2007.
- [14] A. Rényi, “On measures of entropy and information,” *Proceedings of the Fourth Berkeley Symposium*, 1961.
- [15] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. Wiley, 2006.
- [16] M. Stipčević and B. M. Rogina, “Quantum random number generators and their applications in cryptography,” Proceedings of the 2014 International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO), 2014.
- [17] B. Qi, Y.-M. Chi, H.-K. Lo, and L. Qian, “High-speed quantum random number generation by measuring phase noise of a single-mode laser,” *Optics Letters*, vol. 35, no. 3, pp. 312–314, 2010.
- [18] T. Jennewein et al., “A fast and compact quantum random number generator,” *Review of Scientific Instruments*, vol. 71, no. 4, pp. 1675–1680, 2000.
- [19] S. Pironio et al., “Random numbers certified by Bell’s theorem,” *Nature*, vol. 464, pp. 1021–1024, 2010.
- [20] C. Gabriel et al., “A generator for unique quantum random numbers based on vacuum states,” *Nature Photonics*, vol. 4, pp. 711–715, 2010.